Indian Institute of Information Technology Vadodara

CS/IT 429

Behavioral Modeling

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Behavioral Modeling

Procedural Blocks

- initial
- always

Example

- initial
- always
- always@(*)
- always@(a, b, c)
- always@(posedge clk) // 0 to 1, x, z; x to 1, z to 1
- always@(negedge clk) // 1 to 0, x, z; x to 0, z to 0
- always@(posedge clk or negedge clk)
- always@(clk)

Behavioral Modeling

Procedural Blocks

Example

```
module logic1 (i, j, k);
output reg i, j, k;

initial
begin
    i = 0;
    j = 0;
    k = 0;
end

endmodule
```

```
module logic1 (i, j, k);
output reg i, j, k;

initial     i = 0;
initial     j = 0;
initial     k = 0;

endmodule
```

⇩

Identify the output values…!

Behavioral Modeling

Procedural Block

Example

```
module logic2 (i, j, k);
output reg i, j, k;

initial     i = 0;
initial     j = 1;
initial     k = 0;

initial  begin      i = 1; j = 0; k = 1;   end
initial  begin  #1  i = 0; j = 1; k = 0;   end
initial  begin  #3  i = 1; j = 0; k = 1;   end
initial  begin  i = 0;  #4  j = 1; k = 0;  end

endmodule
```

⇨

Identify the output values…!

Behavioral Modeling

Procedural Block

Example

```
module logic3;

initial  #1   $display (" a ");
initial       $display (" b ");
initial  #2   $display (" c ");

initial
begin
   #1   $display (" d ");
        $display (" e ");
   #2   $display (" f ");
end

endmodule
```

⇨ Identify the output values…!

Behavioral Modeling

Procedural Block

Example

```
module logic4 ();
output reg clk1, clk2, clk3, clk4;

initial
begin
  clk1 = 1`b1;   clk2 = 1`b1;
  clk3 = 1`b1;   clk4 = 1`b1;
end

always
  #2 clk1 = ~clk1;

always
  #4 clk2 = ~clk2;

always
begin
  #2 clk3 = ~clk3;
  #4 clk4 = ~clk4;
end

endmodule
```

⇨ Identify the output values…!

Behavioral Modeling

Concept of Procedural Assignments

- Blocking (=)
- Non Blocking (<=)

Behavioral Modeling

Procedural Assignments

Example

```
module logic5 ();
reg clk;
reg [1:0] a1, b1, c1, a2, b2,
c2;

initial
begin
  clk = 1`b1;

  a1 =  2`b00;
  b1 =  2`b01;

  a2 =  2`b00;
  b2 =  2`b01;
end

always
  #1 clk = ~clk;

always@(posedge clk)
begin
   a1 = b1;
   b1 = a1;
end

always@(posedge clk)
begin
   a2 <= b2;
   b2 <= a2;
end

endmodule
```

⇩

Identify the output values…!

Behavioral Modeling

Sequential Circuit Design

- Latch
- Flip-flop

Circuit Designs

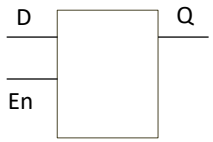
- Synchronous
- Asynchronous

Behavioral Modeling

Design a Latch

Using RTL Model

Using Behavioral Model

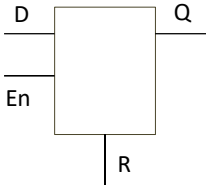


Behavioral Modeling

Design a Asynchronous Latch

Using RTL Model

Using Behavioral Model

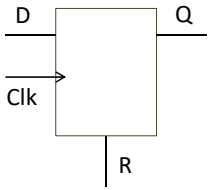


Behavioral Modeling

Design a Flip-flop with Reset

Synchronous

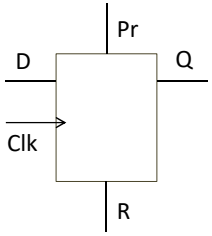
Asynchronous



Design a Flip-flop with Reset-Preset (with Reset highest priority)

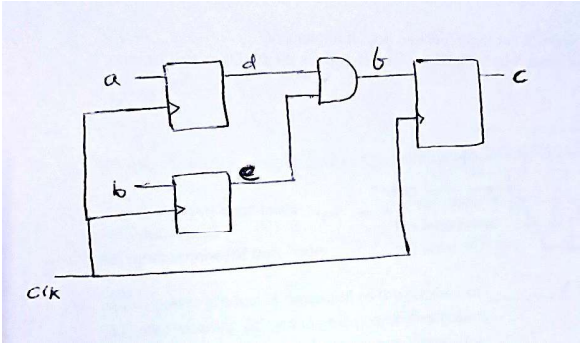
Synchronous

Asynchronous



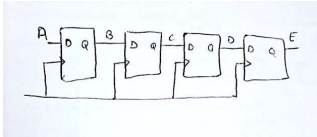
Behavioral Modeling

Implement the System using HDL

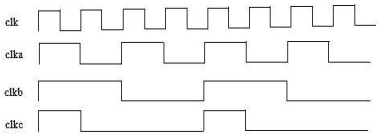


Behavioral Modeling

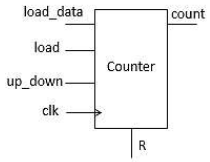
Design a 4-bit Shift Register



Generate the Clock Signals



Design a 3-bit loadable up-down Counter (with Reset)



Behavioral Modeling

Procedural Flow Control

Conditional Statements

- if-else
- case

Loops

- while
- for
- repeat
- forever

Behavioral Modeling

Example

Counter Design using if-else Conditional Statements

Counter Design using case Conditional Statements

Counter number of 1s using while loop

Counter number of 1s using for loop

Usage of repeat loop

Usage of forever loop

I am available/approachable at

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